

NOVOSELENGINSK, Russian Federation

WMO: 308290

Lat: 51.102N

Lon: 106.642E

Elev: 1825

StdP: 13.75

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.5	-28.6	-37.4	0.7	-32.4	-33.5	0.9	-28.2	18.7	8.9	15.9	5.4	1.6	270	0.637

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	88.4	64.4	84.8	63.5	81.3	62.5	67.6	81.9	66.0	79.4	64.6	76.9	7.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.2	93.0	72.3	61.5	87.4	71.0	59.9	82.3	70.2	33.2	82.4	31.9	79.4	30.8	77.1	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	20.3	17.0		-36.4	95.0	5.0	4.6	-40.0	98.3	-43.0	101.0	-45.8	103.5	-49.4	106.8
				-36.6	70.7	5.0	1.8	-40.2	72.0	-43.1	73.1	-45.9	74.1	-49.5	75.4
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.2	-11.7	-3.3	19.0	38.7	51.5	64.7	69.7	64.5	50.6	34.0	12.7	-5.8
	DBStd	29.48	9.65	10.32	12.37	8.18	8.08	7.36	6.03	5.94	7.37	8.01	11.68	10.90
	HDD50	8223	1911	1493	960	349	80	2	0	81	496	1120	1731	
	HDD65	12286	2376	1913	1425	789	424	96	22	82	433	960	1570	2196
	CDD50	1738	0	0	0	9	127	441	610	450	100	1	0	0
	CDD65	329	0	0	0	0	6	86	167	68	2	0	0	0
	CDH74	3260	0	0	0	6	146	968	1511	593	36	0	0	0
	CDH80	1037	0	0	0	0	36	320	521	155	5	0	0	0
	WSAvg	5.8	3.2	3.4	5.9	8.0	8.9	7.4	6.3	5.8	6.5	5.5	4.5	3.5
	PrecAvg	8.7	0.2	0.0	0.0	0.2	0.6	1.4	2.3	2.3	1.0	0.2	0.2	0.2
(15) Precipitation	PrecMax	11.8	0.5	0.2	0.2	0.6	1.4	3.4	4.2	4.3	1.9	0.6	0.5	0.5
	PrecMin	6.3	0.0	0.0	0.0	0.0	0.2	0.3	1.0	0.9	0.3	0.0	0.0	0.0
	PrecStd	1.6	0.1	0.0	0.0	0.2	0.3	0.7	1.0	0.8	0.4	0.2	0.1	0.1
	PrecStd	1.6	0.1	0.0	0.0	0.2	0.3	0.7	1.0	0.8	0.4	0.2	0.1	0.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	18.5	30.8	56.1	71.8	85.5	92.4	94.9	89.8	78.7	61.7	42.8	25.6	
	0.4% MCWB	15.6	25.8	41.6	51.5	59.9	64.3	67.4	65.8	60.7	47.6	34.9	22.9	
	2% DB	11.5	21.7	46.9	64.7	78.2	87.2	89.5	84.0	72.3	55.4	37.1	18.3	
	2% MCWB	9.7	18.3	36.6	47.1	55.9	62.1	66.0	64.0	56.6	43.2	31.2	16.2	
	5% DB	6.6	16.8	42.0	59.5	73.5	83.6	85.6	80.2	67.9	50.5	32.8	13.4	
	5% MCWB	5.1	14.5	33.6	44.5	53.0	61.4	65.2	63.3	54.4	40.8	28.2	11.7	
	10% DB	2.4	12.4	37.2	54.5	68.4	79.6	82.0	76.5	63.6	46.5	29.0	9.6	
	10% MCWB	1.2	10.6	30.5	41.6	50.7	60.0	63.9	62.2	52.1	38.6	25.5	8.2	
	0.4% WB	15.9	25.3	41.9	51.8	60.0	67.5	70.5	68.4	61.6	48.1	35.6	23.1	
	0.4% MCDB	18.4	30.7	55.1	71.2	82.3	84.9	85.8	81.9	76.1	60.3	41.9	25.9	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	9.8	18.6	37.1	47.9	56.8	64.9	68.4	66.2	58.2	44.0	31.6	16.3	
	2% MCDB	11.4	21.7	46.3	63.1	76.9	80.7	83.0	79.6	69.4	53.3	36.5	18.3	
	5% WB	5.2	14.4	33.8	44.9	53.9	63.1	67.1	64.7	55.7	41.4	28.4	11.9	
	5% MCDB	6.5	16.6	41.7	58.8	70.7	78.3	80.8	76.6	65.5	49.6	32.5	13.5	
	10% WB	1.3	10.6	30.6	41.9	51.5	61.5	65.7	63.4	53.2	39.1	25.6	8.3	
	10% MCDB	2.4	12.3	37.0	53.9	66.8	76.2	78.3	74.0	62.4	46.1	29.0	9.6	
(35) Mean Daily Temperature Range	MDBR	16.2	22.7	24.2	22.9	23.3	21.0	18.6	19.0	20.1	20.2	17.3	14.9	
	5% DB	18.6	25.4	27.7	32.4	33.7	28.5	25.0	26.1	27.1	27.1	20.4	16.8	
	5% WB	17.1	22.2	17.7	18.9	16.8	11.1	8.4	10.1	13.1	16.0	15.2	15.2	
	MCDBR	18.4	25.1	26.6	30.8	29.8	23.8	19.7	22.0	23.6	25.6	19.7	16.8	
	MCWBR	16.9	22.0	17.0	18.4	15.3	10.1	7.7	9.6	11.9	15.6	14.9	15.3	
	MCWBR	16.9	22.0	17.0	18.4	15.3	10.1	7.7	9.6	11.9	15.6	14.9	15.3	
(40) Clear-Sky Solar Irradiance	taub	0.269	0.293	0.330	0.378	0.443	0.438	0.429	0.390	0.337	0.310	0.290	0.270	
	taud	2.197	2.181	2.213	2.217	2.050	2.186	2.257	2.351	2.447	2.436	2.289	2.176	
	Ebn at Noon	236	262	273	272	258	260	261	266	271	257	227	213	
	Edh at Noon	23	30	36	40	50	44	41	35	29	23	21	20	
(44) All-Sky Solar Radiation	RadAvg	446	799	1244	1582	1894	1958	1811	1534	1195	761	448	322	
	RadStd	20	30	59	88	99	101	104	83	106	50	27	24	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page